

Cara Yoon

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EDUCATION

• University of British Columbia (UBC)

Anticipated in May 2029

B.Sc. Data Science

Vancouver, BC

- Current Coursework: DSCI 200, STAT 302, STAT 201, MATH 221/111
- Relevant Courses: CPSC 210, CPSC 121, CPSC 110, MATH 200, DSCI 100

PROJECTS

• Java Search Engine | Java, JUnit, Maven, Git

2026



- Designed and built a Java search engine using a positional inverted index to tokenize, index, and retrieve documents from ranked multi-term queries
- Implemented query normalization, term-frequency scoring, deterministic ranking, result limits, and duplicate-document validation
- Developed JUnit tests, structured the project with Maven, and used constructor dependency injection to separate component responsibilities

• BC Wildfire & Air Quality Analytics | R, Quarto, GitHub

2026



- Integrated and audited 2022-2024 Canadian wildfire and BC air quality data, engineering spatial and temporal features to investigate wildfire proximity and daily PM2.5
- Applied exploratory analysis, KNN classification, KNN/linear regression, and K-means clustering to compare air-quality patterns across wildfire conditions
- Published a reproducible Quarto module demonstrating data validation, debugging, and critical auditing of AI-assisted analysis

• Vancouver Crime Analytics | Python, pandas, Jupyter

2026



- Cleaned and analyzed Vancouver open crime data to compare neighbourhood-level crime patterns and offence composition
- Build reproducible visualizations examining crime distribution across neighbourhoods and offence categories

WORK EXPERIENCE

• Python Instructor (Paid)

October 2024 - January 2025

UTG Academy



- Taught Basic-Intermediate Python programming to 4 students (9-14) for 2 hours weekly; each individual completed one project as a curriculum, added personal features, and began an independent project
- Balanced students' ideas with the curriculum and gave constructive, actionable feedback to maintain their motivation

• Robotics Assistant Instructor (Paid)

September 2023 - February 2024

UTG Academy



- Taught VEX Robotics to 12 students ages 9-14 for 8 hours weekly, providing hands-on design, build, and coding lessons
- Transformed students' ideas into achievable designs and adapted demonstrations and feedback to engage

LEADERSHIP EXPERIENCE

• FTC Team Lead

September 2024 - March 2025

PMSS Robotics Club Executive | Port Moody Robotics Team 24232 Lead



- Recruited, onboarded, and led 11-member team of students new to robotics; taught CAD/design and Java programming and established build/code standards
- Managed \$2000 budget and coordinated outreach workshops, including a STEM workshop in an elementary school of 100+ students
- Retained most members with several later advancing into executive roles

SKILLS

- **Programming:** Java, Python, R, C, TypeScript, HTML/CSS
- **Tools:** Git/GitHub, Maven, JUnit, VS Code
- **Data/Analytics:** Python, R, pandas, Jupyter, Quarto, Excel, statistical analysis, data visualization
- **Other:** CAD, Microsoft Excel
- **Languages:** Korean (fluent), Japanese (intermediate)